Spear

FULL CIRCLE RELEASING SPEAR

FISHING

GENERAL DESCRIPTION

INTERNAL-CATCH FISHING SPEAR FOR TUBING, DRILL PIPE AND CASING OF ALL SIZES. FULL CIRCLE SLIPS AND THE TAPERS ON THE BODY SPREAD THE EXPANSION STRAIN OVER A LONG VERTICAL SECTION AND ALMOST THE WHOLE INNER CIRCUMFERENCE OF THE FISH, ALLOWING HEAVY PULLING AND JARRING WITHOUT DISTORTING IT. CAN BE RUN WITH INTERNAL CUTTERS FOR CUT-AND-PULL OPERATIONS.

DRILL PIPE AND TUBING

Spear size	Slips Ø retracted	Slip area (sq in)	Wt (lb)	Body yield (lb)
2-3/8 DP	1-11/16	55	32	56,400
2-3/8 Tbg	1-29/32	59	40	71,500
2-7/8 DP	2	74	55	110,400
2-7/8 Tbg	2-5/16	87	60	134,000
3-1/2 DP	2-7/16	95	71	172,000
3-1/2 Tbg	2-13/16	115	92	224,000
4 DP	3-3/32	126	107	296,000
4 Tbg	3-5/16	135	127	275,000
4-1/2 DP	3-15/32	178	180	318,000
5-1/2 DP	4-1/2	265	270	628,000

CASING

Spear size	Slips Ø retracted	Slip area (sq in)	Wt (lb)	Body yield (lb)
4-1/2	3-25/32	210	132	376,000
5	4	217	169	454,000
5-1/2	4-1/2	265	226	628,000
6	-	315	283	945,000
6-5/8	5-1/2	350	358	1,060,000
7	5-3/4	365	398	1,228,000
7-5/8	6-7/16	430	502	1,414,000
8-5/8	7-1/4	485	652	1,761,000
9-5/8	8-3/16	580	833	2,324,000
10-3/4	9-1/2	690	1,240	3,300,000
11-3/4	10-9/16	785	1,552	4,300,000
13-3/8	12	890	2,017	5,961,000

Dimensions in inches. DP: drill pipe, Tbg: tubing. Yield strengths are theoretical body values (accuracy about 20%), not the strength of the fish; jarring may multiply the pull load by 3 to 10. Each spear has 3 to 6 slips depending on size.



DESIGN FEATURES

- **Full circle slips:** large engagement area, low risk of rupturing the fish.
- **Friction grip:** friction springs up to 3-1/2 in. nominal and 4 in. tubing; friction blocks on larger sizes and casing.
- **Safety ring:** stops the spear entering a fish too small to release from.
- **Two release modes:** bump down and rotate; left-hand wickers on the slips as back-up.
- **Circulation:** safety ring packoff seals inside the fish.

ACCESSORIES

Mill, side hill and sub-type nuts; 36 in. extensions (bored for circulation); skirts and guides for oversize holes; stop subs.

OPERATION

- **Engage:** lower slowly into the fish, rotate at least 1/6 turn to the left, then pull.
- **Release:** bump down full string weight, rotate at least 1/6 turn to the right and raise.
- **Deep or crooked holes:** more rotation may be needed; it does no harm.
- **Always** keep a heavy upward strain when rotating the fish, and bump down before releasing.

HOW TO ORDER

Please specify: 1. spear name and assembly number 2. size and type of top connection 3. size and weight of the pipe to be caught. Recommended spares: 2 sets of slips per fish size, 1 unitizer (or 2 friction block assemblies), 2 friction retainers, 1 stop, 1 safety ring per slip size.